**Week 9: 26th–30th October 2026**

**Subject / الموضوع: Science** **Unit Title / عنوان الوحدة: Unit 1 – Change Over Time – التغير عبر الزمن** **Grade / الصف: 8**

| Lessons الحصص | Learning targets أهداف التعلم | Assessment tools and strategies التقييم Formative تكويني/ Summative ختامي / Differentiation التمايز | Resources and links for Parents و المصادر الروابط المخصصة لأولياء الأمور |
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| **Session 1** الحصة الأولى **Fossils & Earth's History** الأحافير وتاريخ الأرض | **DOK 3 – Analyze:** I can analyze fossil evidence from rock layers and a geologic timeline to explain how life on Earth has changed over time. | 3.1 Activate prior knowledge using organizers, links, and pre-teaching.<br>2.5 Use multiple media to illustrate concepts (diagrams, videos, models).<br>6.3 Facilitate organization with graphic organizers, prompts, and note-taking guides.<br>**Summative:** GRASPS Part 1: Geological Evidence (Criteria A & C) | Kindly check Toddle: Unit 1 GRASPS task PPT and GRASPS Work Sessions PPT (Session 1). |
| **Session 2** الحصة الثانية **Plan & Run Your Investigation** تخطيط الاستقصاء وتنفيذه | **DOK 3 – Design:** I can design and carry out a valid investigation that models natural selection, and justify my hypothesis with scientific reasoning. | 6.2 Support planning with templates, strategy prompts, and modeled thinking.<br>7.1 Increase engagement by offering choices and autonomy.<br>3.3 Guide processing with explicit steps, chunking, and interactive models.<br>**Summative:** GRASPS Part 2, Steps 1–5 (Criterion B) | Kindly check Toddle: GRASPS Work Sessions PPT (Session 2).<br>PhET simulation: phet.colorado.edu/en/simulations/natural-selection |
| **Session 3** الحصة الثالثة **Analyse Results & Build Your Model** تحليل النتائج وبناء النموذج | **DOK 3 – Draw conclusions:** I can analyze my data to draw an evidence-based conclusion and design a model that shows how natural selection changes a population over generations. | 2.2 Clarify syntax/structure and highlight relationships in text, formulas, diagrams.<br>5.3 Build fluency with graduated scaffolds and models.<br>4.1 Provide alternatives for physical action and methods of response.<br>**Summative:** GRASPS Part 2, Steps 6–8 + Part 3 (Criteria C, A & B) | Kindly check Toddle: GRASPS Work Sessions PPT (Session 3). |
| **Session 4** الحصة الرابعة **Conservation Presentation & Submit** عرض توصيات الحفاظ على البيئة والتسليم | **DOK 3 – Construct an argument:** I can use evidence from my portfolio to construct and justify conservation recommendations that protect UAE biodiversity. | 7.2 Make learning relevant, authentic, and connected to students' backgrounds.<br>6.4 Support progress monitoring with rubrics, checklists, and visual progress tools.<br>9.3 Develop self-assessment and self-reflection with tools, charts, and feedback displays. | Kindly check Toddle: GRASPS Work Sessions PPT (Session 4) and the final submission checklist. |

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| | | **Summative:** GRASPS Part 3 finish + Part 4 (Criterion D) | |
| **Homework**<br>**الواجبات** | Anything not finished in class becomes homework before the next session. Students use the "Before you leave" checklist at the end of each session to see what is missing. | 3.4 Support transfer using checklists, organizers, and recurring practice.<br>6.1 Guide goal-setting with prompts, examples, and checklists.<br>8.4 Provide mastery-oriented feedback that emphasizes growth and improvement. | Kindly check Toddle |
| **Teacher's notes/ Reminders**<br>**ملاحظات المعلم والتذكير** | This week students complete their Unit 1 summative performance task (GRASPS): a Scientific Evolution Portfolio in 4 parts. They work as evolutionary biologists for the UAE Biodiversity Research Centre and study a UAE desert mammal in two environments.<br>The full portfolio is submitted at the end of Session 4, and students will present it at the ENS Science Exhibition. Please check with your child that all work is saved in one portfolio on Toddle. | | |

## Week 9 الأسبوع 9

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| **Technology Integration** دمج التكنولوجيا | PhET Natural Selection simulation; Canva, Scratch, stop-motion or animated slides for the evolution model; Toddle for the digital portfolio and submission. |
| **Cross-curricular links** الروابط بين المواد الدراسية | Geography (UAE habitats and Earth's history), Mathematics (graphs, averages and percentage change), Design (building the model) and Individuals and Societies (conservation decisions). |
| **External Assessment Integration** دمج التقييمات الخارجية | Aligned to NGSS MS-ESS1-4, MS-LS4-1, MS-LS4-4 and MS-LS4-6, and MAP Growth Science (Life Sciences). |
| **Wellbeing/ ATL Skills** الصحة النفسية/ أساليب التعلم | Self-management (organising work with checklists and meeting deadlines), research (information literacy and APA referencing), thinking (analysing evidence) and communication (presenting to an audience). |
| **Standards** المعايير | • **MS-ESS1-4:** Construct a scientific explanation based on evidence from rock strata for how the geologic time scale is used to organize Earth's history.<br>• **MS-LS4-1:** Analyze and interpret data for patterns in the fossil record that document the existence, diversity, extinction and change of life forms.<br>• **MS-LS4-4:** Construct an explanation of how genetic variations of traits increase some individuals' probability of surviving and reproducing.<br>• **MS-LS4-6:** Use mathematical representations to support explanations of how natural selection may lead to increases and decreases of traits in populations over time. |
| **Inquiry/ TOK Question** البحث والاستقصاء أسئلة TOK | **Inquiry:** How do environmental changes affect the survival of species over time, and how can science help protect UAE biodiversity?<br>**TOK:** How reliable is fossil evidence as a way of knowing what happened in the distant past? |
| **Success Criteria** معايير النجاح **(Measurable)** | The student has met the learning targets by: لقد حقق الطالب/الطالبة أهداف التعلم من خلال:<br>**Session 1 – Fossils & Earth's History**<br>• **MUST (DOK 1):** I can build a geologic timeline with the 4 eras and at least 6 major events.<br>• **SHOULD (DOK 2):** I can analyse fossils in rock layers and explain which are older or younger.<br>• **COULD (DOK 3):** I can explain how fossils support evolution, using a UAE fossil site as an example.<br>**Session 2 – Plan & Run Your Investigation**<br>• **MUST (DOK 1):** I can write a research question and an If… then… because… hypothesis, and identify my variables.<br>• **SHOULD (DOK 2):** I can write a clear numbered method and collect data in a table with units.<br>• **COULD (DOK 3):** I can justify my hypothesis with scientific reasoning and include repeat trials.<br>**Session 3 – Analyse Results & Build Your Model**<br>• **MUST (DOK 1):** I can draw a correctly labelled graph and describe the trend using numbers.<br>• **SHOULD (DOK 2):** I can explain my results with scientific reasoning, evaluate my method, and start my model. |

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| | • **COULD (DOK 3):** I can discuss the validity of my hypothesis and finish a model that shows all 5 elements.<br>**Session 4 – Conservation Presentation & Submit**<br>• **MUST (DOK 1):** I can finish my model and prepare a presentation showing my evidence and conclusions.<br>• **SHOULD (DOK 2):** I can make a future prediction and a recommendation supported by Claim–Evidence–Reasoning.<br>• **COULD (DOK 3):** I can evaluate the implications of my recommendation (social, environmental, economic, ethical). |
| **National Identity Domain**<br>**مجالات الربط بالهوية الوطنية** | Conservation الحفاظ علي البيئة Global Understanding التفاهم العالمي Heritage التراث |
| | **Area:المجال** Conservation of UAE biodiversity, inspired by Sheikh Zayed's environmental legacy and his protection of the Arabian oryx and UAE wildlife. |
| | **Implementation:التطبيق**<br>• Session 1: students research UAE fossil sites, such as Jebel Hafeet (Al Ain) and the Baynunah fossils (Al Dhafra), and add them to their timeline.<br>• Sessions 2–3: students investigate a UAE desert mammal (such as the Arabian sand gazelle, Arabian tahr or sand cat) in two UAE habitats: sandy dunes (Liwa) and rocky mountains (Hajar Mountains).<br>• Session 4: students recommend ways to protect UAE biodiversity, such as Arabian oryx breeding and reintroduction, the Dubai Desert Conservation Reserve, Sir Bani Yas Island and ghaf tree seed banks. |